Miguel Medina

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EDUCATION

Escuela Superior Politécnica del Litoral (ESPOL) — Guayaquil, Ecuador *Expected Feb 2028*
B.S. in Computer Science Engineering
Relevant Coursework: Data Structures, Object-Oriented Programming, Discrete Mathematics, Database Systems, Software Design, Software Engineering, Systems Programming.
Currently Taking: Computer Organization, Web & Mobile Application Development.
Certification: Michigan English Test — CEFR B2 (University of Michigan, 2023).

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, JavaScript, TypeScript, HTML/CSS.
Frameworks & Libraries: Django, FastAPI, React, React Native, Next.js, JavaFX, LangGraph, Bootstrap, Tailwind CSS.
Tools & Databases: Git, GitHub, Maven, PostgreSQL, Supabase, MySQL, SQLite, REST APIs, Linux.
Concepts: Data Structures & Algorithms, Graph Algorithms, OOP, Multi-Agent Systems, LLM Integration & Guardrails.
Spoken: Spanish (Native), English (Professional Working, B2), French (Basic).

PROJECTS

Brokeate — AI Robo-Advisor — 1st Place, Agentic Scale 2026 (TAWS) *FastAPI · LangGraph · PostgreSQL*
- Won 1st place out of 60+ competing teams at a 48-hour hackathon by building a robo-advisory platform where a LangGraph AI agent explains investment proposals while a database guardrail blocks it from ever inventing a figure.
- Engineered the FastAPI + PostgreSQL backend so every financial number is computed and validated in the database against a closed context before reaching the user, eliminating LLM numeric hallucination.
- Built a human-in-the-loop flow (an advisor approves, edits, or rejects each AI proposal) with a React Native client, and presented the live prototype to a panel of industry judges.

Airport Route Manager — Graph Algorithms Engine *Java 21 · JavaFX · Maven*
- Built an interactive desktop application modeling airports and flight routes as a directed graph (adjacency list), implementing 7+ classic algorithms including BFS, DFS, Dijkstra, Kruskal (MST), and Kosaraju (strongly connected components).
- Optimized shortest-path performance by replacing an O(n) priority-queue update with a lazy-deletion Dijkstra, and implemented Union-Find with path compression and union-by-rank for near-O(1) MST operations.
- Applied the Strategy pattern to switch edge weights (time / distance / cost) at runtime and recompute all weighted algorithms on the fly, rendered on a zoomable, pannable JavaFX canvas.

BiblioRed — Library Management System *Python · Django · Bootstrap 5*
- Developed a full-stack web platform spanning 4 Django apps and 6 relational models (catalog, members, loans) with complete CRUD and a reporting dashboard.
- Implemented user authentication, database migrations, and seed-data fixtures, with a responsive Bootstrap 5 interface.

EXPERIENCE & LEADERSHIP

Hospitality & Operations — Xanterra Travel Collection (Grand Canyon, AZ) *Summers 2024 & 2025*
- Delivered fast-paced front-of-house service (host, food runner, busser) at one of the Grand Canyon's busiest fine-dining restaurants, serving 300+ guests per shift.
- Coordinated directly with kitchen and service teams to sustain throughput and service standards during peak operations in an English-only, multicultural environment.

Hult Prize Entrepreneurship Competition — ESPOL
- Led the technical strategy for a cross-disciplinary team (Computer Science, Nutrition, Biology) to pitch a scalable, tech-enabled business model addressing a local social problem.